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A Retrospective Evaluation Of Team Performance Indicators In Rugby Union

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Abstract

The purpose of this investigation was to demonstrate a method to validate a set of Performance Indicators (PIs). Data from one team (65 matches) was analyzed using 19 PIs categorized into six groups. Frequencies of each PI were recorded at 10- minute time intervals along with match outcomes (win/loss). Odds ratios (OR) derived from logistic regression (>2.0 or <0.5) were used to identify those PIs that contributed to match outcome. Eight PIs were subsequently identified: Ruck Quick had the largest positive effect on match outcome, within the first 20 minutes of the game and also within the 60-70 minute time interval; Territory (Intro 22) had a positive effect on match outcome, its impact was greatest in the first 20 minutes of the match. Two PIs from the Offence group (Gainline + and Gainline +P) had smaller positive effects on match outcome. Ruck Slow had the largest negative effect on winning a match for all time intervals. Other PIs that had negative effects on match outcome included turnover handling, ruck lost and turnover contact. Retrospective match data was used to investigate the efficacy of PIs. This post-hoc confirmation provides a framework for the assessment of PIs in performance analysis.

Keywords: logistic regression; measurement; validity

1. Introduction

In team sports, coaching departments work in conjunction with the analyst to construct a set of Performance Indicators (PIs). These PIs reflect the most important elements of match- play, which may lead to a successful outcome and are used to routinely provide feedback on team performance. In rugby union, a number of researchers (Laird and Lorimer, 2004; Prim *et al.*, 2006; Sasaki *et al.*, 2007) have defined these discrete variables of match- play. Such reports have informed us of the frequency that PIs occur during a match and although this data may be important, it does not provide a clear indication of 'how' these PIs dictate the outcome of the performance and contribute to winning or losing. Furthermore, performance profiles are heavily influenced by the variability exhibited during match- play. This variability in performance is susceptible to performance conditions, for example the opposition, the venue or the time of day. If

we can produce predictive models of situations (performance profiles) that satisfy key purposes of the sport, coaching criteria can be developed to assess exactly how teams win matches and become successful.

The PIs coaches develop are predominantly influenced by individual philosophies (Wright *et al*., 2012). Moreover, coaches are very confident that these PIs they select can be accredited to match success (Wright *et al*, 2012), despite a lack of empirical support for their use. This value placed on PIs by coaches, highlights a need for performance analysts to validate their measures against successful outcomes. Few studies have attempted to validate PIs by comparing between successful and unsuccessful outcomes in rugby union (Hunter and O'Donoghue, 2001; Jones *et al.*, 2004; Vaz *et al.*, 2010). An alternate approach in which to profile performance may be to investigate the patterns and frequencies of specific PIs between successful and unsuccessful matches for one team as a match progresses. By adopting such an approach, it may be possible to build a more meaningful 'profile' of how a team specifically wins over the course of a match. This approach may be utilised to influence future coaching contributions, not just post- match, but by employing a 'live analysis' to educate coaches on what their team could do differently and affect a positive change on match outcome. Indeed, Carling et al. (2005) described a match analysis as the objective recording and examination of behavioural events that occur during competition. To date, despite their wide use within professional rugby union, no match analysis report has assessed potential changes of PIs as a function of time. Therefore, if changes in PIs are analysed as the match advances, it will be possible to assess important time points in addition to PIs. With this information, a performance profile may be constructed to indicate how successful teams elicit PIs during matches. Hence, the purpose of the current investigation is to provide an example from rugby union of how one may validate and demonstrate to the coaching staff the usefulness of a set of PIs in an applied context.

2. Method

2.1. Participants

One professional rugby union team was analysed for a total of 67 matches (35 wins, 30 losses and two draws) spanning two seasons. The two draws were removed from any further analysis due to the low sample size.

2.2. Procedure

All match statistics were devised from a set of coding definitions. These coding definitions explain intricately the activities performed during a match, which represents each PI. Each match used was downloaded from a Database and analysed by three professional Performance Analysts. The analysts have extensive experience within the field and also with using these specific templates and coding definitions.

Each PI was recorded and frequency reports were produced for eight (10 minute) intervals, from the start to finish of the match. Time was taken utilising the precise match clock, which is controlled from 0 to 80 minutes, and controlled for events such as match stoppages and time offs called by the referee.

2.3. Instrumentation

The Sportscode Elite, v. 8.5.2. (Sportstec, Warriewood, New South Wales, Australia) analysis package was used in the current investigation for the identification of pre-defined PIs and the times during the match at which events occurred. The Matrix and Sorter features of Sportscode Elite were used in order to extract match data, which was transferred into an Excel spreadsheet for analysis. Match footage was filmed using professional television production equipment and captured onto a MacBook Pro using Firewire connections. In the majority of the matches filmed, there was more than one camera angle available to the analysts.

2.4. Selection of Key Performance Indicators

The PIs used in this study can be separated into six distinct areas: offense, defence, contact-efficiency, contact, errors and territory. Specifically, these key performance indicators were: *Offense* – Gainline +, Gainline 0, Gainline –, Gainline +P, *Defence* – Tackle +, Tackle 0, Tackle -, Tackle Other, Tackle Missed, *Contact-Efficiency* – Ruck Quick, Ruck Average, Ruck Slow, *Contact* – Ruck Won, Ruck Lost, *Errors* – Turnover Handling, Turnover Contact, Turnover Kicking, Turnover Other and *Territory* – Into 22. These 19 PIs formed the basis of this study.

The development of this specific set of PIs was taken with several steps considered. James *et al.* (2005) had suggested using a three-step approach to the identification of team performance indicators. Two of their steps, the formation of a list of performance indicators and identification of definitions, were not necessary for this study, as this had already been completed. However the identification of which performance indicators to use was an important consideration. In previous research, Vaz *et al.* (2010) found several significant indicators between winning and losing teams in close matches. Ruck & Pass, Mauls Won, Turnovers Won, Passes Completed, Possessions Kicked, Tackles Made & Errors Made were all found to be significantly different. Jones *et al.* (2004) only found two out of their 22 indicators analysed were significantly a function of winning and losing, Lineout Steals and % Of Tries Scored. The findings of positional indicators developed by James *et al.* (2005) appear broad, but they state individual styles of play may mean intra-positional differences could occur. The above findings were taken into consideration and discussed with head coaches (20+ years of elite coaching experience between them) and performance analysts of a professional rugby union team to arrive at and validate a thorough and broad set of key performance indicators. No set-piece indicators were agreed upon as it was felt that the nature of the study would not herald true results of set-piece success, as it was not felt that match time has any significant correlation of the success of set-piece with regards successful performances. An important conclusion was that of the Defence performance indicator area. The four different PIs of Tackling equate to what is effectively a measure of tackle effectiveness. Tackling was both a significant finding of Vaz *et al.* (2010) and was a key positional indicator in James *et al*. (2005) study, as well as being identified by the coaches as an area of significance to them as coaches, but it doesn't appear anywhere on a recent IRB (2010) statistical report of the 2010 Six Nations Championship, meaning the IRB may be missing a crucial link to identifying performances.

2.5. Reliability

Prior to coding, inter-operator reliability was assessed for each PI by two experienced analysts. A sample dataset that included at least 50 occurrences of each PIs was coded by the principal investigator and then re-coded by another analyst who had over 5 years experience of working with the specified coding template. Kappa (Cohen, 1960) and Bowker symmetry ($P<0.05$) were used to assess inter-operator reliability and the findings are reported in Table 1 below. Kappa for all PIs ranged from 0.65 to 1.00, this represents good to perfect levels of agreement (Altman, 1991).

Table 1. Inter-rater reliability of Performance Indicators (PIs)

GROUP	PI	Kappa	Bowker symmetry P-value
Offence	Gainline +	0.84	0.92
	Gainline 0	0.79	0.99
	Gainline -	0.91	0.98
	Gainline +P	1.00	1.00
Defence	Tackle +	0.92	0.17
	Tackle 0	0.82	1.00
	Tackle -	0.65	1.00
	Tackle Other	0.97	1.00
	Tackle Missed	0.88	1.00
Contact efficiency	Ruck Quick	0.83	0.82
	Ruck Average	0.68	1.00
	Ruck Slow	0.86	0.99
Contact	Ruck Won	1.00	1.00
	Ruck Lost	1.00	1.00
Errors	Turnover Contact	1.00	1.00
	Turnover Handling	0.84	0.17
	Turnover kicking	0.97	1.00
	Turnover other	0.97	1.00
Territory	Into 22	0.96	0.92

2.6. Statistical Analysis

Once inter-operator reliability was established, all matches were coded by the principal investigator: frequency of each PI; the time they occurred; and match outcome were recorded as described above. Logistic regression based on a binary match outcome (win/loss) was used to assess the individual effect of each PI within their respective groups for every specified time interval. The obtained odds ratio scores were subsequently used to investigate PI validity. Due to the number of tests performed, p-values for logistic regressions were not considered. Odds ratio, a relative measure of risk, was used to inform us how much more likely that a win would occur when a given PI occurred compared to when it didn't (Morris and Gardner, 1988; Westergren et al., 2001). An odds ratio of ≥ 2.0 was interpreted as a meaningful effect to a winning outcome, whereas an odds ratio of ≤ 0.5 was interpreted as a meaningful effect to losing

a match (Bland and Altman, 2000; Grimes and Schulz, 2008). Odds ratios for PIs of ≥0.5 and ≤2.0 were therefore not considered important to match outcome.

3. Results

From the 19 PIs assessed, four had a positive effect on match outcome (i.e., win) and four had a negative effect on the home team match outcome for a minimum of one of the 10- minute time-intervals. Table 2. shows these PIs (odds ratio of ≥2 or ≤0.5) in order of magnitude (importance).

Table 2. Performance Indicators (PIs) that had an effect on match outcome (positive or negative)

PI	Odds Ratio (95% confidence intervals)
Ruck Quick	4.8 (2.1 to 14.2)
Into 22	4.3 (2.1 to 11.1)
Gainline +	2.7 (1.5 to 5.4)
Gainline +P	2.4 (1.4 to 4.9)
Ruck Slow	0.1 (0.0 to 0.4)
Turnover Handling	0.2 (0.1 to 0.5)
Ruck Lost	0.3 (0.1 to 0.6)
Turnover Contact	0.3 (0.2 to 0.8)

The following Figures (1-5) illustrate the odds ratio for each PI within their respective PI grouping (Offense, Contact-Efficiency, Contact, Errors and Territory).

Figure 1. shows Gainline + had a positive effect to match outcome. In particular, the first and second time intervals were where odds ratios were above 2.0. In addition, Gainline +P had a positive effect to Match outcome at the time intervals of 10-20, 30-40 and 60-70 minutes. The other Offence grouped PIs had little effect.

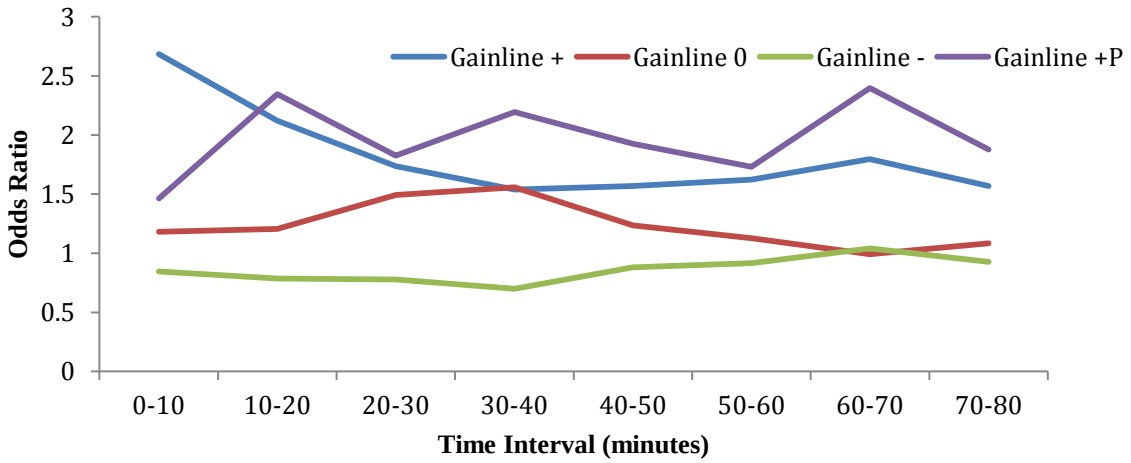

Figure 1. Offense PI Odds Ratios by time intervals for match outcome.

Defence PIs had little effect (odds ratio range from 0.9 to 1.4) at any time interval during the Match. Figure 2 shows the positive effect of Ruck Quick PIs to match outcome was greatest from the start of the game (0-10 minutes) up to 20 minutes and then re-emerged towards the end of the match (60-70 minutes) and the final 10 minutes of the match. In contrast Ruck Slow had a negative effect to match outcome at all time intervals. Ruck Average had little effect at all time intervals.

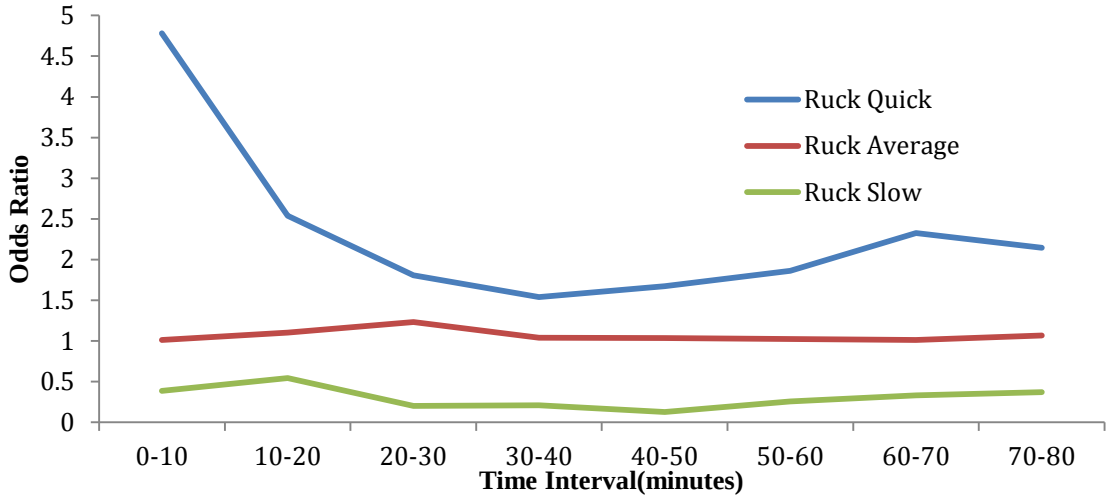

Figure 2. The Odds Ratios of Contact Efficiency PI by time interval

Figure 3 shows that Ruck Lost had a negative effect on match outcome at all time intervals during the first half and at the final 20 minutes of the match. Ruck Won had little effect.

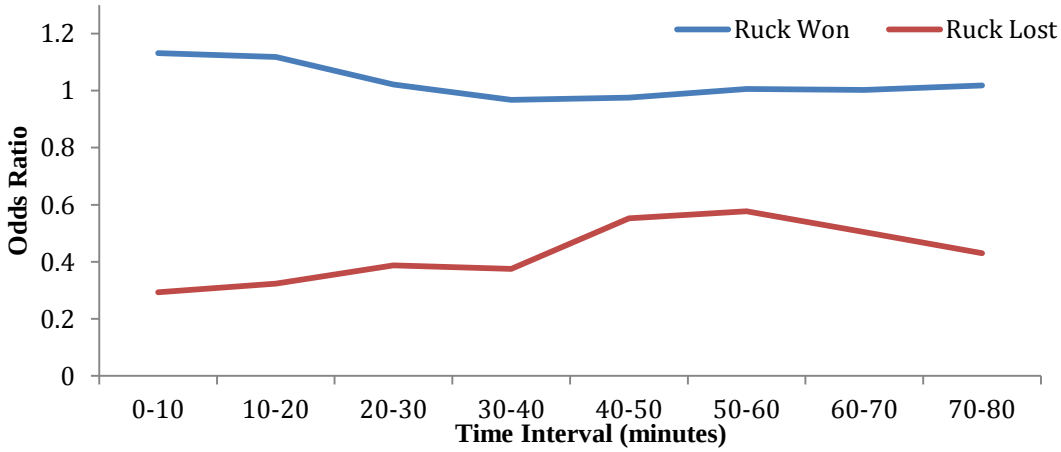

Figure 3. The Odds Ratios of Contact PI by time interval.

Error PI group had a negative effect to match outcome. Shown in Figure 4. Turnover Handling had a negative effect during all time intervals in the second half of the match. Turnover Contact had a negative effect to match outcome in all time intervals in the first half and only in the final time interval in the second half. Turnover Kicking and Turnover Other had little effect on match outcome.

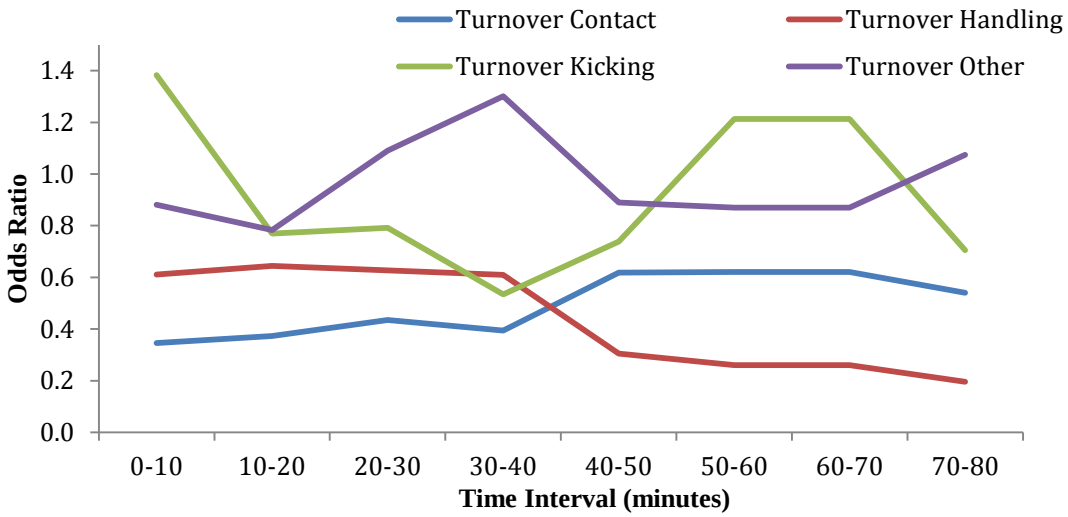

Figure 4. The Odds Ratios of Error PI by time interval.

Figure 5. Shows the importance of territory within the oppositions 22 to winning in particular during the first 20 minutes, 0-10 with odds ratio of 4.2 (95% CI =1.8, 11.1), 10-20 with odds ratio of 4.3 (95% CI =2.1, 11.1), and to a lesser extent over the remaining duration of the match.

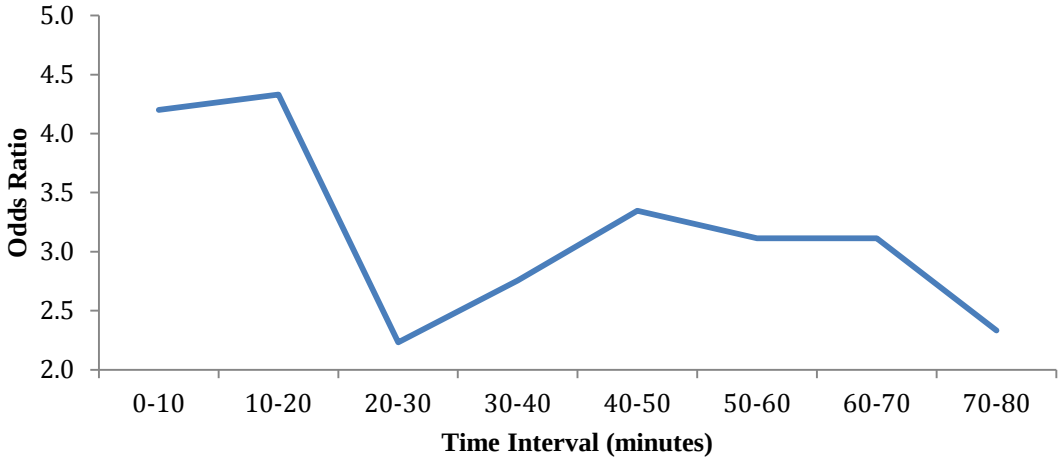

Figure 5. The Odds Ratios of Territory PI by time interval.

4. Discussion

The aim of this investigation was to evaluate a set of PIs using two seasons of match data with known outcomes for one team. Odds ratio (OR) derived from logistic regression was used to assess each group of PIs and their importance to winning the match, throughout the eight equal time intervals. In total, eight PIs were identified contributing to match outcome, as shown in Table 2. Four PIs could be attributed to a successful outcome and four could be attributed to an unsuccessful outcome at specific time intervals. Contact-Efficiency was the most influential group of PIs; with Ruck Quick having a positive effect on match outcome at the first and last 20 minutes of the match, whilst Ruck Slow had a negative effect on match outcome across all time intervals throughout the match.

Results from the current investigation also suggest that Contact Efficiency has a significant effect upon the outcome of a match. The PI, Ruck Slow, proved to be a significant PI throughout the entire match, across all 8-time intervals. Ruck Quick was significant in the first 20 minutes of a match, and also in the 60-70 minute interval. When considering how to breach defences, the slower a ball is at the base of a ruck the greater the amount of time a defence has to reset and plug any potential holes. This slow recycling process could significantly determine a loss throughout the entire match, suggesting that this PI should be considered an important element in the analysis of the match. The slower a team recycles a ball from a ruck in any period of the match, the greater the chance that that team will lose the match. Although this is a very clear indicator, the sport of Rugby Union has stubbornly refuted this. Possibly as a response to the Experimental Law Variations introduced by the IRB (2008), teams became hesitant to be involved in physical 'dog-fights' with opponents. This led to a general trend in rugby union at random times throughout matches to deliberately slow the ball down at the base of a ruck. Teams seemed to believe that this would allow them to 'settle' and begin an attack again safely. The current investigation has suggested that this is not the case, and instead defences are more capable of deflecting these more 'settled' attacks. The IRB has now responded to this global trend (IRB, 2012), which has been accused of boring audiences, by introducing new laws for the speed of a ball at the base of a ruck. Under the new law, if the ball is not recycled within 5 seconds of presentation, then possession will be lost. Only one previous study has considered ruck speed as a performance indicator (Eaves and Hughes, 2003), during this study it was found that there was no significant difference between ruck speeds throughout the four quarters of a match. However, almost 10 years has passed since this finding and the match of Rugby Union will have adapted and changed to incorporate increases in player and match physicality, especially since the sport has turned professional (Cleary and Griffiths, 1997). Further research is required to assess the impact ruck speed has on winning and losing performances, particularly when considering this latest law change.

Domination and crossing the opposition gainline more effectively, particularly in the first 20 minutes of a match, was also found to have a positive effect on match outcome. Similar findings have been reported by Diedrick and Van Rooyen (2011) who noted that the more effectively a team was able to penetrate the opposition defence, the more scoring opportunities were made. Very little work has been done to classify the relationships between points scoring and winning in Rugby Union, therefore, it is

difficult to make inferences, but the IRB (2010) reported that 44% and 59% of tries and penalty kicks were scored respectively in the first half of Six Nations 2010 matches. Despite the small data set, crossing a gainline effectively may lead to more penalty opportunities in the first half, as defences become exposed and threatened. Scoring or scoring opportunities early in a match may also arouse and influence a positive effect for individuals (Kerr and van Sheik, 1985; Cox and Kerr, 1990).

Errors and Turnovers were both found to have a negative effect to match outcome. Turnover Handling was important throughout the second half of matches, as were Turnover Contact and Ruck Lost in the whole of the first half of matches, as well as the time intervals 70-80 and 60-70 respectively. Several other studies (Jones *et al.*, 2004; Van Rooyen *et al.*, 2005; Ortega *et al.*, 2009; Vaz *et al.*, 2011) have all reported that Turnovers are significant in predicting winning and losing performances. Perhaps this is not surprising, given that a Turnover will result in a loss of possession, and points (and ultimately match wins) can only be achieved when a team has possession of the ball. The timings of these turnovers may be important for further study, particularly why ball turned over in contact has a negative effect on match outcome predominantly in the first half, and less so in the second half of matches. Notably, Turnover Handling emerged only in the second half as a negative effect on match outcome. To date, no study has assessed the influence of fatigue on a players handling ability. Gabbett (2008) has, however, assessed the influence of fatigue on tackling ability in Rugby League players, which could be considered to be a closed skill just like passing and catching (although tackling obviously also depends on an opponent). From that study, it was found that the players with the greatest tackling technique in a non-fatigued state had the greatest decrements in tackling ability when placed under fatigued conditions. This may lead to an important coaching consideration, given that the majority of passing and handling drills and exercises tend to occur at the start of a session as warm-ups. Therefore, if players were coached proper handling techniques under fatigued conditions, the probability of a loss may decrease.

The Territory PI of Into 22 was found to successfully predict a win throughout the entire match. Although this may seem quite a simplistic indicator, it provides a method of direct comparison for movement patterns of ones own team and those of the opposition. Furthermore, the more entries by a team and the more they restrict opposition from entering their own 22, the more likely they are to win a match. A more thorough investigation of this territory predictor was provided by Bracewell (2003), who combined discrete individual player actions with the overall teams actions and also investigated team or individual metres gained in certain pitch areas. Furthermore, Jones et al. (2008) investigated match events while in certain territorial areas to depict overall team outcomes. Although both studies found no significance in either area, Garganta *et al.* (1997) provided an example from soccer, indicating that 50% of all goals came from possession obtained in the opponents final third. It was suggested that this was the case due to limited opportunities to tackle and a reduced requirement to pass, leading to a decreased chance of losing possession. Possibly this is why Into 22 had such a strong effect on winning across the entire match, as there are less opportunities for the opposition to affect a scoring opportunity within their own 22.

One unexpected finding was that the Defence group of PIs (tackling effectiveness) had no effect on the match outcome at any time interval. While tackling was shown to be significant in identifying winning and losing performances in previous studies (James *et al.*, 2005; Vaz *et al.*, 2010), it is perhaps reasonable that this was not found in the current study which only investigated one team. At the professional level of Rugby Union few tackles are missed, thus the influence on match outcome is minimal. Future research could look at assessing tackling technique by splitting a tackle into constituent parts, in order to build upon the work of Gabbett et al. (2011) who used the same techniques to determine physiological requirements. This may provide a clearer indication of what determines a successful tackle in a winning performance.

5. Conclusions

As previously discussed, the need for current research to be compatible with what coaches expect to see is becoming more prevalent because of coaches' continuing involvement within PA in their clubs. The aim of this study was to demonstrate how groups of PIs could be evaluated to confirm the importance placed on them by coaching staff as a function of time. The PIs used were developed by the coaching staff based on their coaching philosophy and experience in consultation with the performance analyst. The results from this study were presented as ORs, as such; the order of PI importance within and between PI groups could be assessed and easily conveyed to coaching staff unfamiliar with statistical tests. In addition, the assessment of PIs across 10- minute intervals revealed interesting findings identifying specific windows of opportunity or vulnerability.

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